

Practice Problems

1. Consider the function

$$f(x) = (x + 4)^2.$$

Starting at $x_0 = 3$ and using a learning rate of $\eta = 0.01$, compute the value of x after the first iteration of gradient descent.

- (a) What is the gradient at $x_0 = 3$?
- (b) What is the updated value of x after one step?

2. Suppose you have a dataset with 1000 samples.

- (a) In batch gradient descent, how many samples are used to compute the gradient at each iteration?
- (b) In mini-batch gradient descent with a batch size of 50, how many samples are used per iteration?
- (c) If you perform 10 iterations of mini-batch gradient descent (with replacement allowed between iterations), how many total sample usages occur across all iterations?

3. Consider a binary classification problem with logistic regression hypothesis:

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}.$$

Given parameter vector

$$\theta = \begin{bmatrix} 0.5 \\ -0.25 \end{bmatrix} \quad \text{and input vector} \quad x = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

and the true label $y = 1$,

- (a) Compute $h_{\theta}(x)$.
- (b) Calculate the logistic regression cost for this example.

4. Consider a binary classification problem with feature $x \in \mathbb{R}$ modeled by Gaussian Discriminant Analysis.

The class-conditional densities are:

$$p(x \mid y = 0) = \mathcal{N}(1, 1), \quad p(x \mid y = 1) = \mathcal{N}(3, 1)$$

and the class priors are:

$$P(y = 1) = 0.4, \quad P(y = 0) = 0.6.$$

Given an observed feature value $x = 2$,

- (a) Compute the likelihoods $p(x \mid y = 0)$ and $p(x \mid y = 1)$.
 - (b) Compute the posterior probability $P(y = 1 \mid x = 2)$.
5. Given a dataset with two features: Weather (Sunny, Rainy) and Temperature (Hot, Mild), and a binary class label: Play Tennis (Yes or No). Assume the Naive Bayes classifier assumption that features are conditionally independent given the class.

Weather	Temperature	Play Tennis
Sunny	Hot	No
Sunny	Hot	No
Rainy	Mild	Yes
Sunny	Mild	Yes
Rainy	Hot	Yes

- (a) Explain what Naive Bayes assumes about Weather and Temperature given Play Tennis, and discuss one limitation and one advantage of this assumption.
- (b) Calculate the posterior probabilities

$$P(\text{Play Tennis} = \text{Yes} \mid \text{Weather} = \text{Sunny}, \text{Temperature} = \text{Mild})$$

and

$$P(\text{Play Tennis} = \text{No} \mid \text{Weather} = \text{Sunny}, \text{Temperature} = \text{Mild}).$$

Which class is predicted?

6. (a) Explain the main conceptual difference between bagging and boosting in ensemble learning, focusing on how the base models are trained and combined.
- (b) Describe one advantage and one limitation of each method (bagging and boosting).

7. Consider the multivariate function

$$f(\theta_1, \theta_2) = (\theta_1 - 2)^2 + 4(\theta_2 + 1)^2.$$

Starting at $(\theta_1^{(0)}, \theta_2^{(0)}) = (4, 1)$ with learning rate $\eta = 0.1$:

- (a) Compute the gradient ∇f at the starting point.
 - (b) Perform one step of gradient descent and report the updated parameter values $(\theta_1^{(1)}, \theta_2^{(1)})$.
 - (c) What is the global minimum of f ? Does gradient descent converge to the correct minimum for any positive learning rate? Briefly explain.
8. Consider a logistic regression model with a single feature x and parameters θ_0 (bias) and θ_1 . The model is:

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}.$$

You observe a single training example $(x, y) = (1, 0)$ and current parameters $\theta_0 = 0$, $\theta_1 = 1$.

- (a) Compute the predicted probability $h_\theta(1)$.
 - (b) Compute the binary cross-entropy loss for this example.
 - (c) Write the gradient descent update rule for θ_1 (with learning rate η), and compute the updated θ_1 after one step using $\eta = 0.5$.
9. In logistic regression, the decision boundary is the set of points where $h_\theta(x) = 0.5$.
- (a) Show that the decision boundary corresponds to the hyperplane $\theta^T x = 0$.
 - (b) Suppose $\theta = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}$ (with $x_0 = 1$ as the bias feature). Write the equation of the decision boundary in terms of x_1 and x_2 , and sketch it qualitatively (describe the line in the x_1 - x_2 plane).
 - (c) For a new point $x = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, predict the class label (0 or 1).

10. Consider GDA applied to a two-class problem with n training examples, n_0 from class 0 and n_1 from class 1 (with $n_0 + n_1 = n$). The GDA model assumes:

$$y \sim \text{Bernoulli}(\phi), \quad x | y = k \sim \mathcal{N}(\mu_k, \Sigma),$$

with a shared covariance Σ .

- (a) Write the maximum likelihood estimates for ϕ , μ_0 , and μ_1 in terms of the training data.
- (b) Using your answer above, compute ϕ , μ_0 , μ_1 , and Σ for the following dataset:

$$\text{Class 0: } x \in \{1, 3\}, \quad \text{Class 1: } x \in \{5, 7\}.$$

(Treat each observation as one-dimensional, i.e., $x \in \mathbb{R}$.)

- (c) Classify a new point $x = 4$ using the GDA decision rule. State clearly which class is assigned and why.
11. Consider the following training data for a Naive Bayes spam classifier, where the feature is whether the word “free” appears in the email (Yes/No) and the class is Spam/Not Spam:

“free” appears?	Class
Yes	Spam
Yes	Spam
No	Spam
No	Not Spam
No	Not Spam
Yes	Not Spam

- (a) Estimate (using maximum likelihood) the class prior $P(\text{Spam})$ and the likelihoods $P(\text{“free”} = \text{Yes} | \text{Spam})$ and $P(\text{“free”} = \text{Yes} | \text{Not Spam})$.
- (b) A new email arrives in which “free” appears. Compute $P(\text{Spam} | \text{“free”} = \text{Yes})$ and $P(\text{Not Spam} | \text{“free”} = \text{Yes})$. Which class does Naive Bayes predict?
- (c) Suppose instead that the word “free” appears in zero Spam emails in the training set. Explain the *zero-probability problem* and describe how Laplace smoothing resolves it.

12. Consider building a decision tree to classify the following dataset. The features are Outlook $\in \{\text{Sunny, Overcast}\}$ and Humidity $\in \{\text{High, Normal}\}$, and the label is Play $\in \{\text{Yes, No}\}$:

Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

- (a) Compute the entropy $H(\text{Play})$ of the full dataset.
 - (b) Compute the information gain of splitting on **Outlook** and on **Humidity**.
 - (c) Which feature should be chosen as the root of the decision tree? Justify your answer.
13. A decision tree is grown to full depth on a training set of 100 examples and achieves 100% training accuracy.
- (a) Why might such a fully grown tree perform poorly on unseen test data? What phenomenon does this illustrate?
 - (b) Describe two strategies (other than pruning) to mitigate this issue when using decision trees in practice.
 - (c) Explain how the Random Forest algorithm addresses the over-fitting problem in decision trees, mentioning both *bagging* and *feature randomization*.
14. In AdaBoost, consider a binary classification problem with labels $y_i \in \{-1, +1\}$. After the first round of boosting, suppose you have one weak learner $h_1(x)$ with weighted error $\epsilon_1 = 0.25$.
- (a) Compute the learner weight $\alpha_1 = \frac{1}{2} \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right)$.
 - (b) After computing α_1 , the sample weights are updated. Write the general weight update rule for a misclassified example and a correctly classified example.

- (c) Suppose you have 4 training examples, all with initial weight $\frac{1}{4}$. After round 1, examples 1 and 2 are misclassified and examples 3 and 4 are correctly classified. Compute the new (unnormalized) weights and then normalize them so they sum to 1.
15. Consider gradient descent applied to the function $f(\theta) = \theta^4 - 4\theta^2$.
- (a) Find all stationary points of f (i.e., points where $f'(\theta) = 0$) and classify each as a local minimum, local maximum, or saddle point.
 - (b) Starting from $\theta_0 = 0.1$ with a small learning rate $\eta = 0.05$, describe qualitatively where gradient descent will converge. Would starting from $\theta_0 = -0.1$ lead to a different result? Explain.
 - (c) Explain why non-convex loss surfaces like this one pose a challenge in training machine learning models, and mention one practical strategy to address it.
16. Suppose you are choosing between logistic regression and GDA to classify emails as spam or not, where each email is represented by a bag-of-words feature vector $x \in \mathbb{R}^d$.
- (a) State one key assumption that GDA makes about the data that logistic regression does not. Under what conditions does GDA's assumption hold exactly?
 - (b) If the GDA assumption holds, which classifier is generally expected to perform better with a small training set, and why?
 - (c) In practice, bag-of-words features are often far from Gaussian. Does this favor logistic regression or GDA? Justify your answer in one or two sentences.